

1. Project Overview

This Java-based project is a multi-stage game that begins with a **menu interface**. Upon entering the menu, the user is prompted to enter their **name**. Using this information, the user can either:

- View their previous scores through the **Scoreboard**
- Start a new game by selecting the **Start** option.

2. Game Flow and Mechanics

The game consists of two stages (maps), and both stages are driven by dice rolls.

Stage 1:

- The player rolls a die to move forward on the map.
- Depending on the tile landed on, the player may:
 - **Gain points**
 - **Lose points**
 - **Land on a neutral tile** (no effect)
- Upon reaching the end of the first map, the game proceeds to the second stage.

Stage 2:

- The player continues by rolling the dice.
 - Moving **3 tiles forward**
 - Moving **2 tiles backward**
- The game ends after reaching the end of the second map.

After finishing both stages, the game returns to the **main menu**, where the player can:

- Re-enter their name
- View their **highest score**, **lowest score**, and **all scores** on the **Scoreboard**.

3. Data Structures and File Operations

To manage and store game data, we used the following:

- **Linked List**: Used for tracking score history and gameplay progression.
- **Binary Search Tree (BST)**: Used for efficient score retrieval and sorting, particularly in the Scoreboard feature.
- **File I/O Operations**:
 - Scores are **written to and read from a file**, allowing data persistence between game sessions.
 - When a player views their scores, the program retrieves the data from the file based on the name input.

4. Challenges Faced

- The main challenge occurred during the **second stage of the game**. Specifically, the game would sometimes **end prematurely before the player reached the end of the second map**. Debugging and resolving this required the condition that determines the end of the game

